

Global Facilities – Design & Construction Standard - Mechanical - Cleanroom Makeup Air Unit (CRMAU) System Specification

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1.0 Purpose

- 1.1 The document is intended to supplement the actual Construction Drawings and Construction Specifications for Construction projects.
- 1.2 This Standard is applicable for Cleanroom Makeup Air systems.
- 1.3 The document is intended to supplement the actual Construction Drawings and Construction Specifications for Construction projects.

2.0 Scope

- 2.1 The document covers the new Basebuild System installations only and does NOT cover tool installation.
- 2.2 This document does NOT cover Non-Cleanroom Makeup Air systems.
- 2.3 This document applies to all Micron Facilities employees, contractors, and vendors at all Micron sites for new Greenfield FAB Construction projects.

3.0 Definitions and Acronyms

- **AMC** - Airborne Molecular Contaminants
- **MAU** - Makeup Air Unit for Non cleanroom
- **CRMAU** - Cleanroom Makeup Air Unit
- **CD** - Condensate Drain
- **FAT** - Factory Acceptance Test
- **IST** - Integrated System Test
- **ITP** - Inspection and Test Plan
- **LOTO/COHE** - Lockout Tagout / Control of Hazardous Energy
- **N+1+F** - System sizing concept for equipment quantity consisting of N duty units, +1 redundant standby unit, +1 future unit
- **OAT** - Operational Acceptance Test
- **PDP** - Pressure Dew Point
- **PLC** - Programmable Logic Controller
- **POC/LPOC** - Point of Connection / Lateral Point of Connection
- **SAT** - Site Acceptance Test
- **TUM** - Tool Utility Matrix
- **UPS** - Uninterruptible Power Supply
- **VSD/VFD** - Variable Speed Drive / Variable Frequency Drive

4.0 Roles and Responsibilities

Roles	Responsibilities
Global Facilities Engineering Team - Global (Discipline Engineer)	• Develop and maintain System Standard document • Ensure Best Known Methods (BKM), lessons learned, and emerging technologies are documented in System Standard • Perform periodic document review and update System Standard
Global Facilities Engineering Team - Regional (Discipline Engineer)	• Review and provide technical feedback on System Standard • Support the Site teams by providing knowledge and technical expertise
Site Project Construction Teams	• Review and provide technical feedback on System Standard • Incorporate System Standard into Project Engineering, Design, and Construction • Document project deviations, exceptions, and exclusions from System Standard • Share new lessons learned to Global Facilities Engineering Team, to determine if it is Best Known Method (BKM) that can be captured in future revision of System Standard
Site Facilities Teams	• Use System Standard as technical reference

5.0 Overview of Systems

- 5.1 The Cleanroom Makeup Air system, including cleanroom makeup air unit (CRMAU) and ductwork, is designed for replacement of the process exhaust air volume and the building leakages in the related Cleanroom areas. The CRMAU supply clean, temperature and humidity-controlled air to the cleanroom and maintain the positive pressure of cleanroom.
- 5.2 The Cleanroom Makeup Air Units shall be indoor type and components as follows:
- Pre-Filter
 - Secondary Filter
 - Pre-Heating Coil (if need), Where the climate has the chance with temperature $< 0^{\circ}\text{C}$, need to install Recirculation Pump.
 - Primary Cooling Coil
 - Air Washer (Humidifier)
 - Secondary Cooling Coil(dehumidification) with drop eliminator
 - Re-Heat Coil
 - Fan
 - Chemical Filter
 - HEPA Filter
 - Motorized Shut-Off Damper

Note: If CRMAU has to be located outdoors, it needs to include weatherproofing, roofing, and freeze protection.

6.0 Performance Spec Requirements:

- 6.1 The Cleanroom Makeup Air Unit shall be designed to achieve the operational performance specs as outlined in **Table 1**:

Table 1: Operational Performance Specs for CRMAU

Parameter	Specification Requirement	Remarks
Supply Temperature	12~23°C (53.6 ~ 73.4 °F)	- Measured at outlet of CRMAU - Depending on CR requirement
Supply Dewpoint	8.3 ~ 10.3°C (46.9 ~ 50.5 °F)	Measured at outlet of CRMAU
Cleanroom pressure	Minimum: 7-8 Pa Recommended: 10-12 Pa	Measured at Cleanroom

7.0 Equipment Design and Sizing Requirements:

- 7.1 **Redundancy and Future** – CRMAU shall be designed for N+1+F. All main system equipment shall have N+1 redundancy (with one redundant unit having capability as online standby), and a provision to install one additional future unit (F) for future capacity expansion.
- 7.2 **System Diversity and Sizing Factor** – CRMAU shall be sized so that the N capacity meets 115-120% of the demand at full build (after initial construction of the system), based on the manufacturing tool layout and TUM. Therefore, the CRMAU sizing for N capacity shall have 15-20% spare future capacity over the demand at full build (after initial construction of the system).
- 7.3 To determine the CRMAU demand for N at full build, an approximate diversity factor of 100% of the sum of all exhaust Peak demand from the TUM. The diversity factor can be adjusted via each site's engineering judgement based on previous known diversity factors, or by interpolating a value between the Average and Peak demand from the TUM. In addition, the CRMAU shall be sized to include the capacity of one additional future unit (F). See **Table 2**.

Table 2: CRMAU Sizing

- Cleanroom Makeup Air Requirement D = Total Exhaust inside Cleanroom + Total Pressurization/ Leakage
 1. Total Exhaust inside Cleanroom = Tool Exhaust (from Levels 2, 3, 4) + VMB/GC Exhaust inside Cleanroom + Space/Area Exhaust inside Cleanroom
 2. Total for Pressurization/Leakage = 1 Cleanroom Air Change per hour (based on Cleanroom Volume x 1)
- CRMAU N capacity,
 $N = 115-120\% \times D$
- CRMAU Sizing = $N + 1 + F$

- 7.4 **Equipment Quantity** – CRMAU shall be selected for a minimum of 3 units (2+1) or more, with each unit to be equal in capacity sizing.

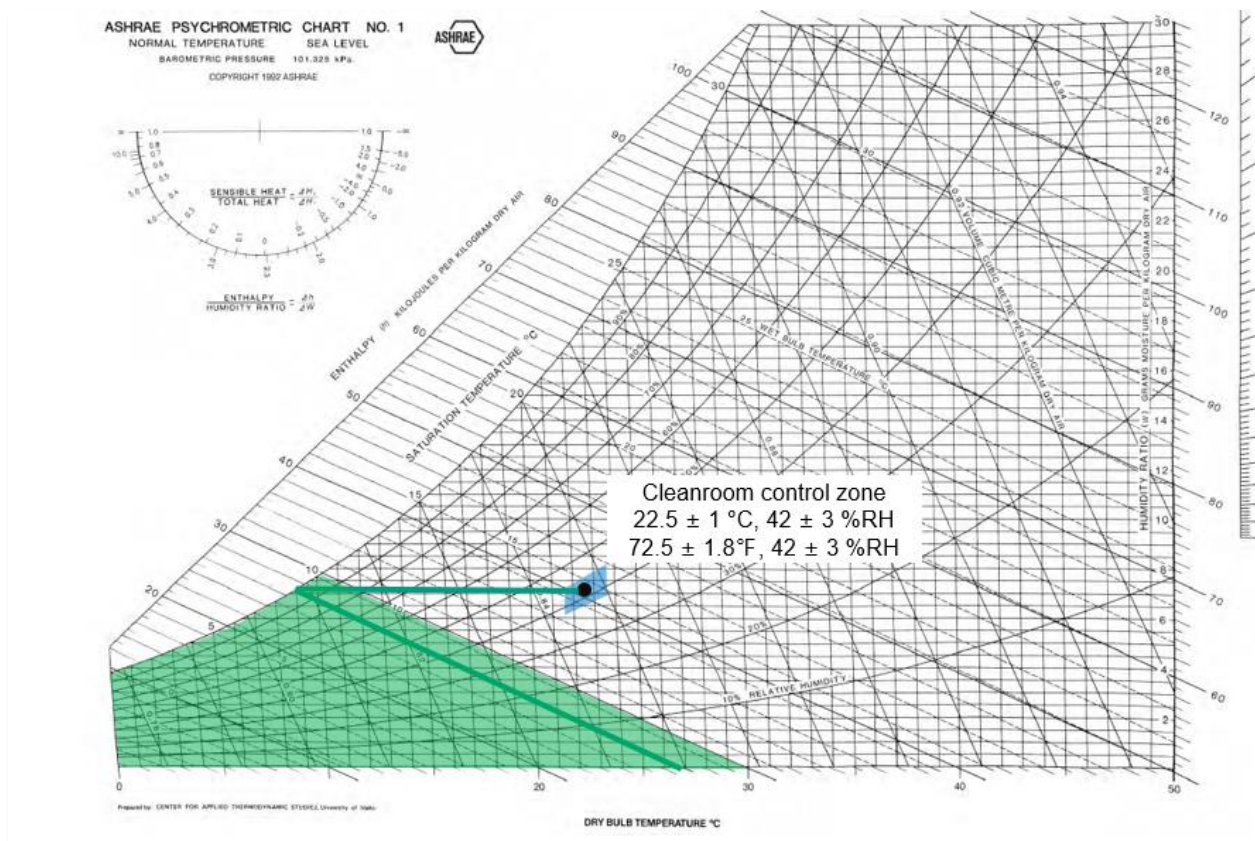
- 7.5 **Equipment Arrangement** – CRMAU shall be arranged in parallel together and connected via a common ducting header for each set. This allows any combination of CRMAU to be in operation concurrently.
- 7.6 **Equipment Layout** – Equipment shall be suitably located to comply with all equipment maintenance clearances and access requirements. Adequate space and a clear routing path shall be reserved for future move-in of new equipment and to allow move-in and move-out of large components during maintenance and repair (such as motors, Fans, Coils etc.).
- 7.7 **Make up air for Photo** – It is recommended that install dedicated CRMAU for Photo area with backup duct from ballroom CRMAU, if possible. Otherwise, the duct with modulating type auto damper to control the pressure of Photo area is required. The modulating type auto damper (in fixed position) shall be adjusted and controlled manually in SCADA (via Operator control). The auto dampers shall not adjust and control automatically.
- 7.8 **Equipment Location**
- 7.8.1 Consider locating and installing the CRMAUs at lower or same floor level of cleanroom, if possible. This is to allow startup of CRMAU and pressurization of the cleanroom as early as possible, during the construction stage. Also, it is recommended to locate the CRMAU indoors, if possible. This is to allow easier operations and maintenance activities on the unit and eliminate the requirements for weatherproofing.
- 7.8.2 Makeup air intakes shall not be placed immediately downstream under prevailing winds of exhaust discharge airstreams **and/or cooling tower discharge airstreams**, in which there is potential risk of cross-contamination of exhaust into the Makeup air intakes **and increase the design cooling load, respectively**. CFD Modeling shall be performed to confirm that there is no cross-contamination of exhaust into the Makeup air intakes.
- 7.9 **Equipment Selection Criteria**
- 7.9.1 The units shall be designed in compliance with the standards for clean room equipment, in particular concerning the tightness of the housing, noise, vibration, stability, corrosion resistance, simple maintenance and continuous operation.
- 7.9.2 The units shall be designed as built-in-place units with frame construction and air tightly installed double walled covering panels of outside painted galvanized steel and inside complete stainless steel, (with glued-on insulating plates) in thermal bridge-free construction.
- 7.9.3 Each unit will rest on a support base frame. In order to avoid any vibrations, special attention shall be paid to the vibration isolation of

the fan and the support frame of the units. Isolation pads shall be installed under the support frame.

7.9.4 Climate considerations –

1. Where the climate has the chance to locate in green area of chart 1(wet bulb temperature < 10°C), must install pre-heating coil in front of air washer for humidification.
2. Where the climate has the chance with temperature < 0°C, must install anti-freeze sensor and interlock to shut off the fan of CRMAU.
3. Where the climate has the chance with temperature < 0°C, pre-heating coil must install recirculation pump to anti-freeze.

Chart 1: climate area for installing pre-heating coil



7.9.5 Coil water source

1. Pre-heating coil: warm water.
2. Primary cooling coil: MTCHW
3. Secondary cooling coil: LTCHW
4. Re-heating coil: warm water. For southeast Asia or similar weather area, should use return line of MT chiller water system as heat source.

7.9.6 All coil sections shall have stainless steel sloped drain pans. From the cooling coil section on the interior shall be out of stainless steel. Face

velocity through the unit approx. 2.5 m/s, through heat exchanger ≤ 2.5 m/s.

- 7.9.7 The units will have one access side for maintenance, repair and exchange of components. All components will be accessible via access doors with windows. The doors will have airtight gasket seals and have latches or hinges.

- 7.9.8 Others selection criteria please See **Table 3**.

Table 3: Equipment Selection Criteria

Equipment	Type and Description
<u>Housing</u>	● Panels for floor, walls and ceiling with a double shell with injected Polyurethane foam; fire resistance of the insulation is B1 according to DIN 4102 part1 ● Both shells made of powder-coated sheet metal ● Surface inside of chamber for filter 1 & 2, pre-heater, pre-cooler, washer, cooler 2 and re-heater to be made of stainless steel (SUS 304) ● Panels must be thermal bridge free and self supporting ● Doors must be built like the panels and fitted with at least 2 hinges and lock bolts with exterior levers, sealing consists of a 2-lip-seal around the doors; doors have to be built with insulation to avoid any condensation on the outside of the unit ● All doors to be equipped with a barrel-lock to be lockable with one single key for all doors and units ● Following unit sections/chambers shall be equipped with doors with visible window and working light with switch outside the chamber: 1. Pre-Filter 2. Secondary Filter 3. Pre-Heating Coil (if required) 4. Primary Cooling Coil 5. Air Washer (Humidifier) 6. Secondary Cooling Coil(dehumidification) with drop eliminator

Equipment	Type and Description
	7. Re-Heat Coil 8. Fan 9. Chemical Filter 10. HEPA Filter ● Provide a test port after air washer for installing on-line AMC sampling tube. ● Housing to be installed on a base frame made of single galvanized steel H-beams which has to fit the different loads of the single chambers of the unit; the base frame to be assembled on site with bolts and nuts to allow the exact alignment of the frame ● For the vibrations elimination will be installed one dedicated layer of pads made of vibration absorbent between the unit housing base frame and the concrete foundation; for the design of the layer of pads a calculation for the critical unit components has to be provided and approved before installation; ● The static pressure test hole shall be provided before and after each component to facilitate the measurement. ● The whole construction has to be suitable for an over- and under-pressure of at least 2,500 Pa ● All housing-parts have to be Silicon-free!
<u>Heating / Cooling Coil</u>	● Frame consisting of stainless sheet steel SUS 304 ● Tubes shall be 5/8-inch OD, seamless continuous copper, without devices that create any turbulence, minimum 0.025-inch wall thickness, serpentine type, with return bends, same-end connections, rolled into headers with copper bushings or brazed into nonferrous headers, arranged for counter flow between air and water. ● Fins shall be minimum 0.0075-inch-thick plate-type aluminum, mechanically bonded to fins and headers. Provide corrugated plate fins to pipes. 1. Provide adequate control of fin spacing and provide a permanent mechanical bond between the pipes and the plate fins, by expanding pipes into fin collars, so that fin collars completely cover the pipes.

Equipment	Type and Description
	● Casings shall be 16-gauge galvanized steel with galvanized steel center tube supports for coils over 42 inches, air bypass seals, and flanged duct connections. ● Headers: 1. Headers shall be manufactured from seamless copper tube. All core joints and brass adopter connections shall be brazed with approved copper brazing alloys. 2. Provide drain plugs to bottom of headers, arranged to fully drain coils. 3. Provide inlet connections at the bottom of supply headers at the "air-off" or "air-on" face of coils (as specified) and outlet connections at the top of return headers at the "air-on" face of coils. ● Fins, tubes and hearer shall be coated with Heresite (wherever specified), epoxy or equivalent to approval of purchaser. ● Fin spacing: minimum 2 mm up to 10 mm according to detailed specification. ● Air Leakage: 1. Minimize air leakage through gaps between the fin ends and frames. 2. Use baffle plates to close the space between the coil frame and the surrounding structure or equipment. 3. Seal gaps between coils and surrounding structure or equipment using non-hardening mastic. ● Piping including drainage and bleeding connection with seal plug. ● Operation pressure for coils 8 bar ● Test pressure minimum 16 bar overpressure, leak test to be done under water according to Pressure Equipment Directive 97/23/EC and EN 378-2000. ● Drain pan 1. Provide drainage from cooling coils and other components where water may collect. 2. The design shall collect condensate and allow complete drainage so that no residue of water remains inside the unit when turned off; extend under the coil both upstream and

Equipment	Type and Description									
	downstream of the coil; extend under droplet separator; and extend under uninsulated piping fittings. The drain trays shall not be flat to avoid residue of water. 3. Material: shall be made of Stainless steel at least SUS 304, > 3mm, in-house welded, with drainage openings. 4. Stacked coils: Provide each coil with a tray and run a minimum DN 40 drain pipe from the upper tray to the lowest tray. Acceptable Manufacturers – Aerofin, Johnson Controls, Huntair, Heatcraft, Trane, Precision Coils, or approved equal									
<u>Droplet Separator</u>	● 2nd Cooler will be equipped with droplet separator consisting of reinforced Polypropylene (PPTV) which allows a continuous operational temperature up to 50° C ● Complete separation of drops at a drop size limit of 15 micron at 2.5 m/s face velocity ● Separation profiles easy to be removed for cleaning and maintenance purposes									
<u>Air Washer</u>	● Washer will be designed to fit the humidification and removing AMC requirements for the Makeup Air supplied to the CR areas. ● Air Washer meets Efficiency ≥ 95% in the process of adiabatic humidification. ● Air washer meets air cleaning efficiency as follows <table><tr><td>AMC</td><td>Inlet concentration range</td><td>Removal Efficiency</td></tr><tr><td>NH₃</td><td>30ppb<Inlet conc.<300ppb</td><td>≥ 90% or ≤ 3ppb</td></tr><tr><td>SO₂</td><td>40ppb<Inlet conc.<400ppb</td><td>≥ 90% or ≤ 3ppb</td></tr></table> ● Washer consist of washer tank, circulation pump, distribution pipes, spray nozzles and Packing plate. ● Packing plate, can keep water on packing’s surface, would increase the liquid-gas contact area and time to improve the AMC removal efficiency and reduce the liquid/gas ratio to save	AMC	Inlet concentration range	Removal Efficiency	NH ₃	30ppb<Inlet conc.<300ppb	≥ 90% or ≤ 3ppb	SO ₂	40ppb<Inlet conc.<400ppb	≥ 90% or ≤ 3ppb
AMC	Inlet concentration range	Removal Efficiency								
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Equipment	Type and Description
	pump energy. The liquid/gas ratio should design <0.2. ● Washer tank shall be design with overflow drain. ● Makeup water design: Two makeup water methods shall be provided in air washer, one is by float valve, and the other is by on-off type control valve. The makeup water quality shall be > 1MΩ at least. ● Washer will be designed as a recirculation humidifier Washer pump ● 2 duty Pump shall be installed outside of the unit, each pump with 100% capacity, total capacity is 2 x 100%. ● Pump manufactured with closed impeller design and labyrinth seal type; connection to the motor by sleeve coupling ● Pump sizing and selection shall be as follows: 1. Pumps shall be selected with the performance indicated at the design operating point (flow, pressure, and motor brake-horsepower). 2. Pump curve shall show the performance indicated at the maximum operating point (flow and pressure, based on pump impeller size, full speed, and full brake-horsepower). 3. Pumps shall be Cast Iron casing, with Bronze, Carbon Steel, or Stainless Steel internally wetted parts (shaft, impeller, etc.). ● Pump motor sizing and selection shall be as follows: 1. Pump motors shall be selected for non-overloading operation on all points of the performance curve at the design conditions. 2. Pump motors shall be rated for inverter duty with variable speed operation. 3. Pump motor efficiency rating shall be minimum IE3.

Equipment	Type and Description
	4. Pump motor IP rating shall be minimum IP55. Spray Nozzles ● High performance atomizing nozzles with hollow cone for full cone distribution ● Large cross section allows high flow rates and avoids nozzle blocking ● Special flat cone allows short dimension of air washer
<u>Fan</u>	● Fans shall be Centrifugal type, Direct-Drive type. ● Fan base frame to be equipped with spring isolators to reduce vibrations ● Fan sizing and selection shall be as follows: 1. Fans shall be selected with the performance indicated at the design operating point (flow and pressure, at partial VSD speed and motor brake-horsepower). 2. Fan curve shall show the performance indicated at the maximum operating point within the fan motor size limitations (flow and pressure, based on fan impeller size, VSD speed, and full brake-horsepower to match motor size rating). Resultant fan tip speed shall be stated. 3. The maximum allowable fan impeller tip speed rating shall be published. 4. The fan maximum operating point tip speed shall be at least 15% - 20% less than the maximum allowable tip speed of the fan. Meaning, for continuous operation at the fan maximum operating point, the fan impeller tip speed shall be less than 80% - 85% of the maximum allowable fan impeller tip speed rating. 5. Fan impeller shall be AMCA rated, Class IV 6. Heavy duty bearings 7. Shaft shall have critical speed 125% of fans maximum operating speed 8. Mechanical efficiency shall be 75% above. 9. Fan assembly shall be statically and dynamically balanced.

Equipment	Type and Description						
	10. Totally enclosed fan-cooled motor with over-temperature protection. 11. Sound level performance to be < 75 dBA at distance of 1m away from fan enclosure. - Fan motor sizing and selection shall be as follows: - Fan motors shall be selected for non-overloading operation on all points of the performance curve at the design conditions. - Fan motors shall be rated for inverter duty with variable speed operation. - Fan motor efficiency rating shall be minimum IE3. - Fan motor IP rating shall be minimum IP55. Acceptable Manufacturers – New York Blower (NYB), Twin City Fan (TCF), or approved equivalent						
<u>Filter</u>	- Filter frames shall be stainless steel with spring fasteners out of stainless steel. The inside flanges shall be gasketed for positive seal. - Pre-filter: rated at 25 to 35% (G4, MERV 7) Recommend pressure drop: 65/130 Pa (clean/dirty) - Secondary filter: rated at 85-95% (F7-F8, MERV 13-14) Recommend pressure drop: 150/250 Pa (clean/dirty) - Chemical filter: 2 racks, to allow 2 different types of chemical filters to be installed for eliminate O₃, H₂S and Total acid (Include F⁻, Cl⁻, Br⁻, NO₃⁻, SO₄²⁻, PO₄³⁻ and NO₂⁻). - It recommended that Chemical filter should meet the required outlet performance as below: <table border="1" data-bbox="751 1476 1128 1614"> <tr> <th>AMC</th><th>Removal Efficiency</th></tr> <tr> <td>H₂S</td><td>≤ 3ppb</td></tr> <tr> <td>O₃</td><td>≤ 3ppb</td></tr> </table> (Refer to OMT & Design & Construction (<5ppb) Standard requirement) - HEPA filter: rated at 99.95% (HEPA H13) Recommend pressure drop: 250/500 Pa (clean/dirty)	AMC	Removal Efficiency	H ₂ S	≤ 3ppb	O ₃	≤ 3ppb
AMC	Removal Efficiency						
H ₂ S	≤ 3ppb						
O ₃	≤ 3ppb						
<u>Outlet Motorized</u>	Multi blades damper: Frame consisting of rolled galvanized sheet steel C-profiles with slots in all corners to facilitate assembly with adjoining parts; frame assembled						

Equipment	Type and Description
<u>Shut Off Damper</u>	without any seam welding ● Blades made of extruded aluminum sheath as hollow profiles with slim and streamline shaped section to give maximum rigidity against distortion and bending ● Bearings for the blades made maintenance-free, consisting of oil impregnated sintered bronze bearings ● Rod-linkage with minimum tolerances for installation of external actuator, including mounting base plate for actuator; no operating elements in the air stream

8.0 Duct Design and Sizing Requirements:

8.1 Ductwork Diversity and Sizing Factor – The CRMAU ductwork shall be sized according to the same sizing factors (N+F, System Diversity, etc.) stated in the above section 7.3 and below.

8.2 Ductwork Mains Sizing Calculation – CRMAU Mains shall be sized to accommodate 115-120% of the demand at full build (after initial construction of the system), based on the manufacturing tool layout and TUM, plus the Facilities Basebuild Requirements. Therefore, the Mains shall have 15-20% spare future capacity over the demand at full build after initial construction of the system). In addition, the Mains shall be sized to include the capacity of future units. See **Table 4**.

Table 4: CRMAU Duct Mains Sizing Calculation

● CRMAU Main Demand at Full Build, D: $D = \text{Total Exhaust inside Cleanroom} + \text{Total Pressurization/ Leakage}$ 1. Total Exhaust inside Cleanroom = Tool Exhaust (from Levels 2,3, 4) + VMB/GC Exhaust inside Cleanroom + Space/Area Exhaust inside Cleanroom 2. Total for Pressurization/Leakage = 1 Cleanroom Air Change per hour (based on Cleanroom Volume x 1) ● CRMAU System N capacity: $N = 115-120\% \times D$ ● CRMAU Main Duct Sizing to include future units: $\text{Sizing} = N + 1 + F$ CRMAU load to be split evenly across Main Ductwork
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8.3 Duct Sizing Criteria – Main Duct Sizing shall be selected based on the full build capacity, including the future equipment (N+1+F). It is

recommended for the duct sizing to be designed with maximum specific pressure drop not to exceed 1.0 Pa/m.

- 8.3.1 The ductwork system pressure drop calculations shall demonstrate that the CRMAU fan selections have adequate performance rating to meet the design parameters and system requirements.
- 8.3.2 It recommended that the external static pressure of CRMAU should be 500Pa at least.
- 8.4 **Duct Pressure Rating, Mains** – CRMAU ductwork mains shall be designed to meet the minimum pressure class of +6 inches w.g. (+1,500 Pa). If the system pressure at the main ductwork exceeds this amount, then the ductwork pressure rating shall meet or exceed the system pressure.
- 8.5 **Duct Pressure Rating, Submains/Laterals** – CRMAU ductwork laterals and submains shall be designed to meet a pressure class of +4 inches w.g. (+1,000 Pa). If the system pressure at the submain/lateral ductwork exceeds this amount, then the ductwork pressure rating shall meet or exceed the system pressure.
- 8.6 **Materials of Construction** – Ductwork, rectangular and circular, shall be made of Galvanized Steel. Ductwork shall be insulated from the CRMAU unit outlet connection up to the discharge opening in RA plenum /truss / CR plenum area, and in all unconditioned spaces. The ductwork insulation material shall be vapor-tight, with a closed cell structure, made of expanded cellular polyethylene foam.
- **Gaskets:** Neoprene tape, silicon-free gaskets shall be used for CRMAU Ductwork.
- 8.7 **Ductwork Installation** - Low pressure rectangular and circular ductwork shall be made of galvanized steel sheet. The supply air ducts shall be thoroughly cleaned before installation and the ends shall be capped during transportation. The ductwork system shall be complete with transitions, bends, tees, supports, dampers, offsets, flexible connections, take-offs and similar fittings necessary for the balancing and full operation of the air distribution system. Ductwork shall be installed in accordance with SMACNA "Industrial Duct Construction Standards."
- 8.8 **Duct distribution** – For uniform distributed supply air, it is recommended to branch ductwork evenly with submain supply the air to return air chase. Each submain shall install modulating type auto damper (in fixed position) for air balancing and adjustment to be made via SCADA (manually by Operator control). Makeup air supply location is recommended at the middle of Return air chase to provide sufficient mixing of makeup air and recirculation air, to prevent temperature and humidity stratification and imbalance. The modulating type auto dampers (in fixed position) shall be installed at ducts supplying to dedicated pressure control area, for pressure control. The monitoring status and the control function of the auto dampers shall be via SCADA (manually by

Operator control). The auto dampers shall not adjust and control automatically.

8.9 Provisions for Main Expansion

8.9.1 At locations in which there is a planned future extension of the main ducts, headers, or laterals/sub-laterals, provide full-size dampers with capped blind flanges at the ductwork ends, to allow easier extension without interruption to the system operation.

8.9.2 At locations of unplanned future extension, provide full-size blind flanges at the ductwork ends.

8.10 Spare Dampers for Future Laterals/Submains - Additional spare dampers shall be included on the main ductwork, to allow installation of future laterals/submains for capacity expansion.

9.0 Accessories:

9.1 The following accessories which shall be installed: See **Table 5**.

Table 5: Accessories to be installed

Accessories	Type and Installation Location	Remarks
Insulation	• Install for all CHW piping throughout the system • Install for all MAU Supply Air ductwork in unconditioned spaces • Install for drain piping	In accordance with Spec Section 15260 (or equivalent Spec)
Piping Supports	• Install piping hangers, supports, racks, and clamps at the appropriate spacing interval as required by the system and piping load and layout • Provide seismic restraints and supports as deemed required during design, based on site location and local jurisdiction requirements • Provide vibration isolation for piping supports as deemed required during design, based on the building vibration criteria	In accordance with Spec Section 15140 (or equivalent Spec)
Ductwork Supports	• Provide duct hanger and support material gauges in compliance with details on the Drawings. Where details are not provided, comply with SMACNA Industrial Duct Construction Standards. • Ductwork hanger and suspension material may be threaded rod, structural shapes, or strut material. The use of wire as suspension material is prohibited.	In accordance with Spec Section 15140 (or equivalent Spec)
Pressure Gauges	• Install at cooler supply and return connection for CHW • Install at Heater supply and return connection for WW • Install at washer pump discharge	
Pressure Difference Gauges	• Install at each filter section	

Accessories	Type and Installation Location	Remarks
Temperature Gauges	• Install at cooler supply and return connection for CHW • Install at Heater supply and return connection for WW	
Air Vents	• Air vents shall be minimum size DN20 (3/4 inches) • Install Auto Air Vents at all high point locations for CHW and Warm water.	
Drain Ports	• Install at all low point locations	
Valves	• All butterfly valves shall be rated for “bi-directional” duty with positive shutoff in both directions • All butterfly valves shall be lug-style	In accordance with Spec Section 15101 (or equivalent Spec)
Control Valves	• Provide control valve type and quantity as outlined in this standard	
Labeling and Identification	• Provide identification labels at minimum of every 6m distance and change of direction • Refer to the specific project’s Construction Specification, (i.e. Spec Section 15190 or equivalent Spec) for additional requirements.	In accordance with Spec Section 15190 (or equivalent Spec)

10.0 Sustainability Design:

- 10.1 **CRMAU Parallel Header Design** – CRMAU shall be arranged in parallel and connected to a common ductwork header. This allows multiple CRMAU to operate together in parallel, to lower the fan operating speed and fan energy consumption.
- 10.2 **Ductwork Pressure Drop Sizing** – It is recommended to increase the ductwork sizes where possible, to minimize the air velocity and pressure drop through the ductwork. This will result in lower CRMAU fan energy consumption.
- 10.3 **Condensate Water reclaim and re-use** – There are 2 types of CRMAU condensate which can be recycled or reuse. The condensate of 1st cooling coil is blended into diluted acid wastewater into AWR system. The condensate of 2nd cooling coil sent to Cooling Tower makeup tank for direct reuse.
- 10.4 **Heat recovery design** – For enhance the overall energy efficiency, the heat source of warm water, for pre-heating and re-heating coil, should recover from MT chiller and/or compressor, rather than Boiler. For Fab in Southeast Asia or similar weather area, the heat source of re-heating coil should design with the return line of MT chiller water system.
- 10.5 **Combined Coil Design** – For the CRMAU required pre-heating coil, it is recommended that take one dual function coil, with heating and cooling, instead of one pre-heating coil plus one cooling coil, due to Pre-heating coil and 1st cooling will not work at the same time. With this kind of design, it can save the initial cost and running cost because of less pressure drop.

11.0 Electrical Design:

- 11.1 **Power Segregation** – Electrical power supply for CRMAU shall be segregated among a minimum of 2 separate upstream power sources (including panel + transformer). The quantity of equipment served from each upstream power source shall be split evenly.
- 11.2 **Un-interrupted Power for Controls** – 100% of CRMAU controls including PLCs, Local Control Panels (LCPs), Central Controller, CRMAU Control Power and other auxiliary control power shall be connected to an Uninterrupted Power source (such as UPS or DC Bank) to allow continuous operation of the CRMAU during momentary power interruption or voltage sags, spikes, and fluctuation events.
- 11.3 **Emergency Power** – A minimum of 50% of CRMAU shall be connected to Emergency Power.
- This is per requirements of Factory Mutual (FM) Standard 1056, National Fire Protection Agency (NFPA) Standard 318, and International Building Code (IBC) Section 415, to maintain 50% capacity (exhaust airflow) upon a power loss.
- The interpretation is that by having 50% of the exhaust fans on emergency power, it is sufficient to provide approximately ~50% capacity (exhaust airflow) upon a power loss, to satisfy the requirements.
- To maintain proper cleanroom pressurization levels, then a minimum 50% of Cleanroom Makeup Air Units shall also be on emergency power.
- 11.4 **Variable Frequency Drives** – CRMAU fans shall be VFD driven for variable speed operation. Variable Frequency Drives shall be selected to be one size larger than the power rating of the actual CRMAU fan motor size.
- 11.4.1 Variable Frequency Drives shall be compliant with SEMI standard F47 for voltage sag immunity, to provide ride-through capability during momentary power interruption or voltage sag.
- 11.4.2 VFD should have auto-reset function after power outage. The startup sequency should be controlled by FMCS.
- 11.5 **Local Disconnect Switches** – Each CRMAU shall have a lockable type, local disconnect switch installed adjacent to CRMAU, to allow for manual isolation and lockout/tagout (LOTO/COHE) of the washer pumps and fan.
- 11.6 **Air washer pump** – The power source of air washer pump should be the same with CRMAU fan.

12.0 Instrumentation:

12.1 The following Instrumentation (at least but not limited) shall be installed and monitored/ controlled via FMCS. See **Table 6**.

Table 6: Instrumentation to be installed

Instrumentation	Type and Installation Location
Pressure Transmitters	Install at Main header (Quantity: 1 duty and 1 backup [2 Total])
Delta Pressure Transmitters	- Install at each filter bank (Quantity: 1 duty)
Temperature/Humidity Transmitters	- Install at outlet of CRMAU (Quantity: 1 duty) - Can be convert to Dewpoint & Enthalpy
Temperature Transmitter	Install at outlet of heater and Cooler (Quantity: 1 duty)
Conductivity Transmitter	Install at Air washer tank (Quantity: 1 duty)
Level Transmitters	Install at Air washer tank (Quantity: 1 duty)

13.0 Control Sequence Philosophy:

13.1 The overall philosophy for system control and operation shall be listed in this section. See **Table 7**. Refer to the specific Sequence of Operations for each project for specific details.

13.2 Air Washer pump/valve on/off status, conductivity, water level, alarms, and pump/valve on/off control shall be shown and operated at the local control panel, and connected back to SCADA for monitoring and control.

Table 7: System Control Logic

Control Parameter	Control Logic
CRMAU Pressure Control	- Speed output of CRMAU Fan VSDs shall be modulated to maintain the pressure setpoint at the Main duct. - The pressure setpoint shall be averaged across the number of pressure-sensing locations spread across the Main duct.
CRMAU Dewpoint Control	Dehumidification - Position output of 1st cooler control valve shall be modulated to maintain the temperature setpoint at outlet of 1st cooler. - Position output of 2nd cooler control valve shall be modulated to maintain the Dewpoint setpoint at outlet of CRMAU.

Control Parameter	Control Logic
	Humidification - Position output of Pre-heater and 1st cooler control valves shall be modulated to maintain the Dewpoint setpoint at outlet of CRMAU. - Pre-heater and 1st cooler control valves <u>must not</u> be opened at the same time.
CRMAU Temperature Control	Position output of Re-heater control valve shall be modulated to maintain the setpoint at outlet of CRMAU.
Air washer Blowdown and Makeup Water Control	- Air washer conductivity blowdown and makeup water control shall be automatically done via the CRMAU's integral control algorithm via the Local Control Panel. - Upon sensing high conductivity levels, the blowdown drain valve shall open to begin blowdown. - Upon sensing low level, blowdown drain valve shall close and the makeup water valve shall open to begin makeup. - It recommended to blowdown at 19µS/cm and stop at 15µS/cm. The setpoint can be adjust according to different conditions.
CRMAU Start/Stop	CRMAU Start/Stop shall be done automatically via SCADA based on required number of running units

13.3 CRMAU Fan VFD shall be failed last speed.

13.4 Washer pumps shall be interlock with CRMAU Fan; in case both (2) washer pumps fail, CRMAU fan shall shut down and send alarm to SCADA.

13.5 The following table shows the control valve fail positions that shall be configured. See Table 8.

Table 8: Control Valve Fail Positions

Control Valve	Control Valve Fail Position		
	Loss of Air	Loss of Electricity	Loss of Control Signal
CRMAU pre-heating coil control valve	Fail Last State	Fail Last State	Fail Last State
CRMAU pre-heating coil control valve (the climate has the chance with temperature < 0°C)	Fail open State	Fail open State	Fail open State
CRMAU 1 st cooling coil control valve	Fail Last State	Fail Last State	Fail Last State
CRMAU 1 st cooling coil control valve (the climate has the chance with temperature < 0°C)	Fail open State	Fail open State	Fail open State
CRMAU 2 nd cooling coil control valve	Fail Last State	Fail Last State	Fail Last State

Control Valve	Control Valve Fail Position		
	Loss of Air	Loss of Electricity	Loss of Control Signal
CRMAU Re-heating coil control valve	Fail Last State	Fail Last State	Fail Last State
Air washer backup water control valve	Fail Closed	Fail Closed	Fail Closed
Air washer blowdown control valve	Fail Closed	Fail Closed	Fail Closed
Discharge control dampers	Fail Last State	Fail Last State	Fail Last State
Submain modulating damper	Fail Last State	Fail Last State	Fail Last State

- 13.6 Any control valve/damper actuators configured to fail-last state shall have the capability of failing to last state position and holding the position for unlimited duration without needing control air, power, or PLC signal.
- 13.7 Upon a loss event of control air, power, or PLC signal, and then after the utility is restored, the control valve/damper output command/position shall be set to the same command/position it was operating at prior to the loss event.
- 13.8 All control valves/dampers shall have a manual override bypass, to allow manual operation of the valve/damper upon a failure.

14.0 Smart Facilities Design:

- 14.1 AMC Online Monitoring – It required to install AMC online monitoring system to allow continuous monitoring, O₃, H₂S, NH₃, Acid substance (HF&HCL), and alarm of air washer and chemical filter status.
- 14.2 The design of fan vibration measurement should be able to take from outside of the CRMAU rather than going into Fan chamber to install the device and take measurement.
- 14.3 CRMAU housing should install a small maintenance door for placement of AMC impingers to avoid using normal maintenance door. The location of impinger maintenance door should be able to check the efficiency of air washer and each chemical filter.
- 14.4 The following are additional enhancements that can be considered and included to enable Smart Facilities Design concepts.
 - 14.4.1 **CRMAU Fan Vibration Online Monitoring** – Consider installing vibration sensors on CRMAU fan and motor, to allow continuous monitoring and alarm of vibration levels.
 - 14.4.2 **CRMAU Dewpoint cascade control design** – Consider take the dewpoint in cleanroom to reset the dewpoint setpoint of CRMAU, to stabilize the cleanroom dewpoint.
 - 1. For sites with relatively constant climate year-round (i.e. Singapore, Malaysia), either fixed control setpoint or Auto-Cascade control setpoint can be considered for the Makeup Air discharge dewpoint controls.
 - 2. For sites with seasonal climate changes for summer, winter, fall, and spring, (i.e. USA, Japan, Taiwan, China), it is recommended to use Auto-Cascade control capability to automatically adjust the Makeup Air Discharge Dewpoint setpoint.
 - a. Upper and Lower limit clamps shall be set to prevent Makeup Air Discharge Dewpoint from drifting excessively from the normal setpoint.
 - b. Upper and Lower alarm limits shall be set to notify Operations if the Makeup Air Discharge Dewpoint drifts excessively from the normal setpoint.
 - c. Either fixed control setpoint or Auto-Cascade control setpoint can be choice for dewpoint controls.

15.0 Inter-Discipline Coordination Matrix:

15.1 The following table shows the provisions that shall be provided by each of the following disciplines and coordinated accordingly. See **Table 9**.

Table 9: Provisions Provided by Each Discipline

Discipline	Provision
Architectural	• Provide dedicated Utility Room or Mechanical Room space for centralized CRMAU • Provide concrete housekeeping pad below CRMAU • Provide floor drains as required in Utility Rooms • Cut openings in walls and floors for piping and ductwork penetrations, and seal openings after piping and ductwork installation • Provide adequate fire stopping at all required fire wall penetrations
Mechanical	• Provide HVAC space conditioning for dedicated Utility Room or Mechanical Room space containing the centralized CRMAU. Room temperature shall be designed to maintain the space temperature range of 24°C to 28°C. • Provide heating water, chilled water and city water supply piping, and accessories as outlined in this Standard, to provide sufficient heating, cooling and city water for CRMAU • Provide drain water and blowdown water piping, and accessories as outlined in this Standard • Provide floor drains located suitably to allow proper draining of equipment, condensate drains and auto-drains, piping, and air vents
Electrical	• Provide VFDs for all CRMAU fans • Provide local disconnect switches for MAU • Provide power wiring for all Electrical equipment/components • Provide grounding for all Electrical equipment/components • Provide Emergency Power / UPS Power as outlined in this Standard • Provide Power Segregation as outlined in this Standard
Water / Wastewater	• Provide DI Water piping for Air Washer • Provide drain source for the Coil Drain water piping and air washer blowdown water piping.

Discipline	Provision
Controls	• Provide control valve actuators and positioners • Provide control wiring • Provide instrument control air tubing • Provide FMCS/SCADA system monitoring and controls of all instrumentation and transmitters including pressure, temperature, dewpoint, flow, etc. • Provide Zero Reference Pipe (ZRP) for pressure reference • Integrate CRMAU Central Controller with FMCS/SCADA

16.0 Quality Inspection and Testing:

16.1 **Inspection and Test Plan (ITP)** – Create formal Inspection and Test Plans (ITP) that outline all requirements for quality verification, inspections, testing, and commissioning:

16.1.1 Piping and Ductwork

- Piping and Ductwork Shop Drawings and Submittals
- Incoming Material Inspection
- Material Storage
- Welding Procedures, Qualifications, and Inspection
- Piping, Ductwork, and Supports Installation and Verification
- QA/QC Inspection & Punchlist
- Leak Testing and Pressure Testing
- Cleaning, Flushing, and Purging
- Quality Certification
- Final QA/QC Inspection & Punchlist
- Training, Documentation, and Handover

16.1.2 Equipment

- Equipment Shop Drawings and Submittals
- Factory Acceptance Test
- Incoming Equipment Inspection
- Equipment Move-In and Installation
- QA/QC Inspection & Punchlist
- Equipment Pre-Startup Checks – Mechanical, Electrical, Controls, etc.
- Operational Acceptance Test (OAT) - Standalone Equipment Startup per Vendor's Procedure to allow equipment to run in MANUAL mode. Also considered "Early Run" Equipment.
- Final QA/QC Inspection & Punchlist
- Training, Documentation, and Handover

16.1.3 System

- Site Acceptance Test (SAT) – Test System operation and control functions in accordance with P&ID, SOO, and Specifications.
- Integrated System Test (IST) – Test full System operation including all system interdependencies, interlocks, and critical failure modes.
- Final QA/QC Inspection & Punchlist
- Training, Documentation, and Handover

16.2 **QA/QC Inspection** – Prior to leak testing and pressure testing, QA/QC inspection shall be performed on all installed piping, ducting and components to verify proper installation in accordance with construction drawings, project specifications, and industry standards.

16.3 **Quality Certification Test** – HEPA Filter leak testing, filter frame leak testing shall be performed, to meet the Quality Certification requirements outlined in this standard.

17.0 Startup Commissioning / Site Acceptance Test

17.1 At a minimum, the following system control functions shall be tested and successfully proven during the System Startup Commissioning / Site Acceptance Test.

17.1.1 Fans

- Fan Motor Rotation Check
- Fan/Motor Alignment Check and Lubrication
- All Damper, Valve, and Actuator Checks
- Motorized Damper operation checks including actuator, shaft, linkages, and connection points
- VSD Settings and Functionality Checks
- Control interface and communication checks to FMCS/SCADA
- Fan Standalone Startup Procedure per Manufacturer's procedure
- Fan Vibration Level Checks
- Verify Fans Operation through its operating range (0 to 50/60 Hz)
- Verify Fan Start/Stop Function to Rotate Operating Fan
- Verify Fan Auto-Start sequencing, upon failure of an operating Fan
- Verify Fan Speed control in response to CRMAU header duct and/or Cleanroom pressure fluctuations
- Verify Motorized Damper opens upon fan startup and closes

- upon fan stop, while keeping supply pressure stable
- Confirm cleanroom makeup air system pressure control is stable during fan manual start/stop rotation
- Confirm cleanroom makeup air system pressure control is stable during fan fault/trip event and next fan auto-start sequencing
- Verify VSD fails last speed upon any failure
- Verify control interface and communication checks to FMCS/SCADA

17.1.2 Air washer

- Pump Motor Rotation Check
- Pump/Motor Alignment Check and Lubrication
- VSD Settings and Functionality Checks
- Control interface and communication checks to FMCS/SCADA
- Pump Standalone Startup Procedure per Manufacturer's procedure
- Verify Pump Operation through its operating range (0 to 50/60 Hz)
- Verify Pump Rotation Start/Stop Function
- Verify Pump Auto-Start sequencing, upon failure of an operating Pump
- All Valve, Actuator, switch and level sensor Checks
- Control interface and communication checks to Local Control Panel, PLC, and FMCS/SCADA
- Confirm supply water pressure in the range per Manufacturer's suggestion during pump running.
- Check the spray pattern and clogging of spray nozzles.
- Verify blowdown drain and makeup sequencing and control

17.1.3 Integrated system performance test

- Confirm system pressure control is stable during fan manual start/stop rotation, and fan auto-start procedure
- Confirm Auto-switch to backup pressure transmitter(s) upon failure of main pressure transmitter(s)
- Confirm all control valves and dampers fail to the stated position per this Standard, upon loss of instrument air, power, and/or control signal.
- Verify all alarms are enabled and activated in SCADA
- Verify equipment on emergency power is able to operate on emergency power, after simulated loss of normal power
- Perform Comprehensive Power loss and recovery (blackout) test

17.1.4 Instrumentation and Controls

- Confirm Auto-switch to backup main header pressure transmitter upon failure of main pressure transmitter
- Confirm all control valves/dampers fail to the stated position per this Standard, upon loss of instrument air, power, and/or control signal.
- Verify all alarms are enabled and activated in SCADA

18.0 Reference Specifications:

18.1 The following Construction Specifications shall be referenced during design, engineering, and construction. The Specifications shown below are in accordance with the Construction Specifications Institute (CSI) Master Format; Refer to the specific Construction Specifications for each project. See **Table 10**.

Table 10: Reference Construction Specifications

Spec Section	Spec Contents	Remarks
Section 15140	Piping Supports and Anchors	
Section 15190	Piping Labels and Identification	
Section 15260	Piping Insulation	
Section 15280	Equipment Insulation	
Section 15515	Hydronic Specialties	
Section 15990	Testing, Adjusting and Balancing	
Section 15170	Motors	
Section 15172	Variable Frequency Drives	
Section 15070	Stainless Steel Piping	
Section 15101	Valves	
Section 15540	Pumps	
Section 11223	Filters	
Section 15242	Vibration Isolation	

19.0 Document Control

19.1 Approval:

GLOBAL_FAC_GFTT_DOC_APPR

19.2 Notification:

GLOBAL_FAC_MANAGERS
GLOBAL_FAC_COMMUNICATIONS
GLOBAL_FAC_CONSTRUCTION
GLOBAL_FAC_NOTIFY
FCG_F4_FAC_NOTIFY
FCG_F6_FAC_NOTIFY
FCG_F10_FAC_NOTIFY
FCG_F11_FAC_NOTIFY
FCG_F15_FAC_NOTIFY
FCG_F16_FAC_NOTIFY
FCG_MMP_FAC_NOTIFY
FCG_MMY_FAC_NOTIFY
FCG_MSB_FAC_NOTIFY
FCG_MTB_FAC_NOTIFY
FCG_MXA_FAC_NOTIFY

19.3 Final Viewer Access

All Micron Team Members from all Micron Sites

19.4 Retention

Adherence to the requirements specified in the
[**Global Facilities - Document Control Procedures**](#)

19.5 Review

Biennially (two years)

20.0 Revision History

Revision #	Section Changed	Details of Changes	Date
0	NA	New document	03/26/21
1	Section 3	3.0 Added FCT, GFTT, GFC, Global Facilities Central Team, Global Facilities Construction Team, Global Facilities Operations Support Team, Global Facilities Planning Team	03/14/23
	Section 5	5.2 iii Updated.	
	Section 7	7.9.4 Item 3 Added 7.9.5 Item 4 Updated Table 3 Updated	
	Section 10	10.4 Updated	
	Section 11	11.4.2 Added 11.4.6 Added	
	Section 14	14.2 Added 14.3 Added	

2	3.0	Remove Acronym of FCT, GFC, GFTT, Global Facilities Central Team, Construction Team, Operations Support Team and planning Team.	05/02/24
	Table 4	Modify Roles and Responsibilities table with Global Facilities Engineering Team – Global, Regional and site Projection construction Teams	
	7.8.2	Add “and/or cooling tower discharge airstreams” and “and increase the design cooling load, respectively.”	
	Table 8	Add control valve status of CRMAU pre-heating coil control valve (the climate has the chance with temperature < 0°C) And CRMAU 1st cooling coil control valve (the climate has the chance with temperature < 0°C)	

Appendix 1: Template Form for Site Specific Deviation List

Site: _____

No	Specification Section and Item	Deviation Proposal	Scope Impact (Benefit, Cost, Schedule)	Remark	Approval: 1. Site 2. GFTT